

Ghanashyam Bharadwaz Avula

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SUMMARY

Computer Engineering graduate with experience in building production grade AI applications, combining software engineering with LLMs to deliver scalable, real-world solutions. Take a closer look at my work in my [Portfolio](#). Looking for backend or AI engineering roles where I can ship production systems. Certified by Anthropic and IBM in Generative AI and machine learning.

EDUCATION

Sreenidhi Institute of Science and Technology (Hyderabad, IN)

Bachelor of Technology in Computer Engineering

2022 – 2026

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS

Frameworks: React, Node.js, FastAPI

Databases & DevOps: PostgreSQL, MongoDB, ChromaDB, Redis

CI/CD & Cloud: Git, Docker, AWS

Gen AI: LangChain, Transformers, RAG, MCP, Vector Search, LLM APIs

ML: TensorFlow, Scikit-learn, Matplotlib, Seaborn

PROJECTS

QuizForge - Live Quiz Platform | *React, Node.js, ChromaDB, HuggingFace, Redis*

[\[Live Link\]](#)

- Built a live quiz platform that auto-generates questions from uploaded course docs using RAG, supporting 50+ concurrent users with real-time leaderboards..
- Implemented MERN stack with JWT authentication, LangChain RAG pipeline with ChromaDB vector search, HuggingFace LLM, and Redis caching layer.
- Containerized the app with context-grounded quiz generation with anti-hallucination guardrail system and reduced API token consumption through intelligent caching.

GATE Assistant | *React, FastAPI, PostgreSQL, WebSockets, Mistral LLM, REST API*

[\[Live Link\]](#)

- Deployed a full-stack AI study planner for GATE exam with personalized daily schedules, revision timelines, and a study-buddy matching feature based on plan similarity scoring.
- Leveraged REST API architecture for JSON extraction and parsing, WebSockets and Redis for real-time updates and caching, and HuggingFace Mistral model with a fallback LLM strategy.
- Onboarded more than 40 users with features like automatic model switching on failure and peer recommendations with aligned preparation strategies.

AskMyDB - NL to SQL Engine | *LangChain, Llama LLM, Graph RAG, SQLite*

[\[Live Link\]](#)

- Orchestrated a natural language to SQL pipeline with hybrid retrieval, multi-layer safety guardrails, and zero-config database onboarding for plain English to optimized SQL conversion.
- Combined Vector RAG + Graph RAG for retrieval which helped in establishing relations between tables. Then implemented LangChain pipeline with Mistral LLM and write-operation interception.
- Passed 31 out of 34 test cases including edge cases in accurate SQL conversion across multi-table joins, schema edge cases, zero unintended destructive queries.

Fine-Tuning LLMs for Fake News Detection | *TensorFlow, HuggingFace, DistilBERT, Python*

[\[Github\]](#)

- Conducted research on automated fake news identification, resulting in a supervised learning pipeline that employs fine-tuning of transformers for fake news classification.
- Utilized TensorFlow, Huggingface tokenization pipeline and DistilBERT transfer learning, for retraining the top few layers of the model while freezing the rest.
- Achieved 92% classification accuracy with minimal training data and compute, delivering production-ready inference from a pretrained foundation model.

EXPERIENCE

AICTE Google AI-ML Virtual Internship

June 2024 – August 2024

Google / AICTE

Remote

- Worked on integrating multiple image classification models using TensorFlow and Keras for ‘Breast cancer detection’.
- Improved F1 score from 0.78 to 0.92 on a dataset containing over 2000 images.

ACHIEVEMENTS & CERTIFICATIONS

- Participated in Smart India Hackathon 2023 as a backend developer, collaborating with my team on “Smart Crop Management System for Farmers” project.
- Claude code using Google Cloud Vertex AI– Anthropic
- IBM Machine Learning Specialization – Coursera
- Secured 97 percentile in GATE 2026.